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A Stochastic Differential Equation Framework for Multi-Objective LLM Interactions: Dynamical Systems Analysis with Code Generation Applications

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Abstract

We introduce a general stochastic differential equation framework for modeling multi-objective optimization dynamics in iterative Large Language Model (LLM) interactions. Our framework captures the inherent stochasticity of LLM responses through explicit diffusion terms and reveals systematic interference patterns between competing objectives via an interference matrix formulation. We validate our theoretical framework using iterative code generation as a proof-of-concept application, analyzing 400 sessions across security, efficiency, and functionality objectives. Our results demonstrate strategy-dependent convergence behaviors with rates ranging from 0.33 to 1.29, and predictive accuracy achieving $R^2 = 0.74$ for balanced approaches. This work proposes the feasibility of dynamical systems analysis for multi-objective LLM interactions, with code generation serving as an initial validation domain.

1. Introduction

Iterative interactions with Large Language Models across multiple objectives present fundamental challenges in dynamical systems theory. As LLMs become integral to complex decision-making processes, from content generation to reasoning tasks, understanding how competing objectives evolve through successive interactions becomes crucial for algorithm design and system optimization. We introduce a stochastic differential equation (SDE) framework that captures the inherent randomness and multi-objective trade-offs in LLM interactions. Our approach models the continuous-time evolution of objective vectors through drift-diffusion processes, enabling rigorous analysis of convergence properties, stability conditions, and interference patterns between competing goals.

The theoretical insights enable systematic design of interaction strategies that achieve desired convergence properties. To support our framework, we apply it to iterative code generation, a domain where security, efficiency, and functionality objectives naturally compete. However, the mathematical foundations could extend broadly to any multi-objective LLM application, including content optimization, reasoning enhancement, and human-AI collaboration systems.

Our contributions support dynamical systems theory as a foundation for understanding and optimizing multi-objective LLM interactions, with immediate applications to algorithm design and broader implications for AI system optimization.

2. Related Work

Classical work by Robbins and Monro Robbins and Monro (1951) established stochastic approximation theory, while modern extensions by Borkar Borkar (2009) and Dieuleveut et al. Dieuleveut et al. (2020) address non-convex settings. Our framework extends these foundations to multi-objective LLM interactions, introducing the interference matrix concept for objective coupling analysis.

Traditional multi-objective optimization Deb et al. (2002); Coello et al. (2007) assumes deterministic objective functions and focuses on Pareto-optimal solutions. Recent work on multi-objective LLM systems Zhang et al. (2021) explores neural architecture search, while Liu et al. Liu et al. (2024) examine multi-objective alignment from human feedback. Our stochastic differential equation approach uniquely addresses the inherent randomness in LLM responses and dynamic objective evolution through interactions.

Emerging research investigates LLM optimization through various lenses. Language-Model-Based Evolutionary Optimizer (LEO) Ma et al. (2024) applies population-based strategies, while recent work on LLM cascades with multi-objective considerations addresses performance-cost-privacy trade-offs Liu et al. (2024). However, these approaches lack the mathematical rigor of dynamical systems analysis for understanding convergence properties and interference patterns.

Our framework provides a systematic dynamical systems foundation for multi-objective LLM interactions, enabling principled algorithm design and theoretical analysis of convergence behaviors across diverse applications. The research on human-AI collaboration in software development includes studies by Vaithilingam et al. Vaithilingam et al. (2022) on programmer expectations and Barke et al. Barke et al. (2023) on grounded copilot usage. However, systematic analysis of objective trade-offs in iterative collaboration remains unexplored.

3. Stochastic Dynamical Systems Framework for Multi-Objective LLM Interactions

3.1. General Mathematical Formulation

Consider an iterative LLM system optimizing n competing objectives. Let $\mathbf{x}^{(t)} \in \mathbb{R}^n$ represent the objective vector at iteration t. We model the continuous-time evolution as a stochastic differential equation:

$$d\mathbf{x} = \boldsymbol{\mu}(\mathbf{x}, \pi)dt + \boldsymbol{\sigma}(\mathbf{x}, \pi)d\mathbf{W} \tag{1}$$

where $\boldsymbol{\mu}(\mathbf{x}, \pi) : \mathbb{R}^n \times \Pi \to \mathbb{R}^n$ is the drift vector encoding systematic objective changes under strategy $\pi \in \Pi$, $\boldsymbol{\sigma}(\mathbf{x}, \pi) : \mathbb{R}^n \times \Pi \to \mathbb{R}^{n \times n}$ captures LLM response variability, and $\mathbf{W}$ is an n-dimensional Brownian motion.

This formulation provides a general framework for analyzing any multi-objective LLM interaction, from content generation with quality-diversity trade-offs to reasoning systems balancing accuracy and efficiency.

3.2. Theoretical Foundation for SDE Modeling

We model discrete LLM interactions using stochastic differential equations based on Euler-Maruyama approximation theory. Consider the discrete process with unit time steps:

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)} + \boldsymbol{\mu}(\mathbf{x}^{(t)})\Delta t + \boldsymbol{\sigma}\sqrt{\Delta t}\boldsymbol{\varepsilon}^{(t)} \tag{2}$$

where $\boldsymbol{\varepsilon}^{(t)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ represents standardized LLM response variability and $\Delta t = 1$ represents the iteration interval.

This corresponds to the Euler-Maruyama discretization of the SDE:

$$d\mathbf{x} = \boldsymbol{\mu}(\mathbf{x})dt + \boldsymbol{\sigma}d\mathbf{W} \tag{3}$$

For unit steps ($\Delta t = 1$), the systems exhibit:

Matching moments: $\mathbb{E}[\Delta\mathbf{x}|\mathbf{x}] = \boldsymbol{\mu}(\mathbf{x})$ and $\mathrm{Cov}[\Delta\mathbf{x}|\mathbf{x}] = \boldsymbol{\sigma}\boldsymbol{\sigma}^T$, Related eigenvalues: $\lambda_{discrete} = 1 + \lambda_{continuous}\Delta t$ for linearized drift matrices, Different stability criteria: continuous stability ($\mathrm{Re}(\lambda_{continuous}) < 0$) corresponds to discrete stability ($|\lambda_{discrete}| < 1$), Asymptotically consistent dynamics: qualitative behaviors (convergence patterns, oscillations) align between formulations

While invariant distributions may differ due to finite-step effects, the SDE framework provides the natural mathematical foundation for analyzing convergence properties and objective trade-offs in discrete LLM interactions.

3.3. Interference Matrix and Objective Coupling

We define the **interference matrix $\mathbf{I} \in \mathbb{R}^{n\times n}$** with off-diagonal elements quantifying cross-objective correlations:

$$I_{ij} = \begin{cases} \mathrm{Corr}(\Delta x_i^{(t)}, \Delta x_j^{(t)}) & \text{if } i \neq j \\ 0 & \text{if } i = j \end{cases} \tag{4}$$

where $\Delta x_i^{(t)} = x_i^{(t+1)} - x_i^{(t)}$ represents the change in objective i. By convention, diagonal elements are set to zero to emphasize cross-objective interference patterns. Negative off-diagonal elements indicate systematic trade-offs between objectives.

For linear SDE systems, these correlations capture the composite effects of: (1) systematic coupling through the drift matrix $\mathbf{A}$, (2) noise correlations from the diffusion matrix $\boldsymbol{\Sigma}$, (3) transient dynamics from initial conditions, and (4) time-averaging effects across the trajectory. Rather than measuring isolated causal mechanisms, our interference matrix characterizes the net empirical coupling experienced by multi-objective LLM systems in practice. This composite measure reflects the tendency for objectives to move together or in opposition. The interference matrix provides a general tool for analyzing multi-objective dynamics across diverse LLM applications like content generation, reasoning tasks and dialogue systems.

3.4. Dynamical Regimes and Eigenvalue Analysis

For the linearized system $d\mathbf{x} = \mathbf{A}\mathbf{x}dt + \boldsymbol{\Sigma}d\mathbf{W}$ near equilibrium, the eigenvalue spectrum of $\mathbf{A}$ determines convergence behavior: